Dataset and benchmark limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete source-grouping and benchmark-coverage record.

Keep row identities, weights, slice denominators, exclusions and evidence gaps together.

Keep one habit: Recall → W1–W3/W5 → independent source-count case → research project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Sensitivity analysis

> **My remembered idea, example and correction**

Multiple comparisons

> **My remembered idea, example and correction**

Uncertainty intervals

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A benchmark measures its particular sources, tasks, labels and conditions. Main ten rows include three copies of one source and seven other distinct source IDs: eight distinct sources, with two extra rows beyond one per source. The repeated source has three times the row weight of each other source. Different IDs do not prove independent sampling.

  Preserve records and declare the unit/exclusion rule; no per-source outcomes are supplied for a deduplicated accuracy. Separate picture: nine of ten clean indoor photos are correct, giving 90% indoors. No outdoor photos are initially present, so outdoor performance is unknown, not zero. Changed scenario supplies four reserved outdoor cases, two correct: 50% outdoors.

  Pooling gives 11/14, approximately 78.6%, which must not replace the outdoor slice. The small slices do not establish future representativeness, cause or significance. Inspect eligibility, source dates, labels and grouping. No actual model run or photo collection is performed by the local metadata arithmetic.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- A benchmark measures its particular sources, tasks, labels and conditions. Main ten rows include three copies of one source and seven other distinct source IDs: eight distinct sources, with two extra rows beyond one per source. The repeated source has three times the row weight of each other source. Different IDs do not prove independent sampling.

  Preserve records and declare the unit/exclusion rule; no per-source outcomes are supplied for a deduplicated accuracy. Separate picture: nine of ten clean indoor photos are correct, giving 90% indoors. No outdoor photos are initially present, so outdoor performance is unknown, not zero. Changed scenario supplies four reserved outdoor cases, two correct: 50% outdoors.

  Pooling gives 11/14, approximately 78.6%, which must not replace the outdoor slice. The small slices do not establish future representativeness, cause or significance. Inspect eligibility, source dates, labels and grouping. No actual model run or photo collection is performed by the local metadata arithmetic.

- Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

Declared source ID	Row IDs in illustrative arrangement	Row weight
S1	R1, R2, R3	3 of 10
S2	R4	1 of 10
S3	R5	1 of 10
S4	R6	1 of 10
S5	R7	1 of 10
S6	R8	1 of 10

Declared source ID	Row IDs in illustrative arrangement	Row weight
S7	R9	1 of 10
S8	R10	1 of 10
Total	Ten rows; eight distinct IDs	S1 receives triple row weight
Evidence limit	IDs illustrate supplied arrangement	No labels or independent sampling inferred

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A benchmark measures its particular sources, tasks, labels and conditions. Main ten rows include three copies of one source and seven other distinct source IDs: eight distinct sources, with two extra rows beyond one per source. The repeated source has three times the row weight of each other source. Different IDs do not prove independent sampling.

  Preserve records and declare the unit/exclusion rule; no per-source outcomes are supplied for a deduplicated accuracy. Separate picture: nine of ten clean indoor photos are correct, giving 90% indoors. No outdoor photos are initially present, so outdoor performance is unknown, not zero. Changed scenario supplies four reserved outdoor cases, two correct: 50% outdoors.

  Pooling gives 11/14, approximately 78.6%, which must not replace the outdoor slice. The small slices do not establish future representativeness, cause or significance. Inspect eligibility, source dates, labels and grouping. No actual model run or photo collection is performed by the local metadata arithmetic.

- A report omits this qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A benchmark measures its particular sources, tasks, labels and conditions. Main ten rows include three copies of one source and seven other distinct source IDs: eight distinct sources, with two extra rows beyond one per source. The repeated source has three times the row weight of each other source. Different IDs do not prove independent sampling.

  Preserve records and declare the unit/exclusion rule; no per-source outcomes are supplied for a deduplicated accuracy. Separate picture: nine of ten clean indoor photos are correct, giving 90% indoors. No outdoor photos are initially present, so outdoor performance is unknown, not zero. Changed scenario supplies four reserved outdoor cases, two correct: 50% outdoors.

  Pooling gives 11/14, approximately 78.6%, which must not replace the outdoor slice. The small slices do not establish future representativeness, cause or significance. Inspect eligibility, source dates, labels and grouping. No actual model run or photo collection is performed by the local metadata arithmetic.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled dataset and **benchmark** limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of dataset and **benchmark** limitations.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Dataset and **benchmark** limitations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Dataset and **benchmark** limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

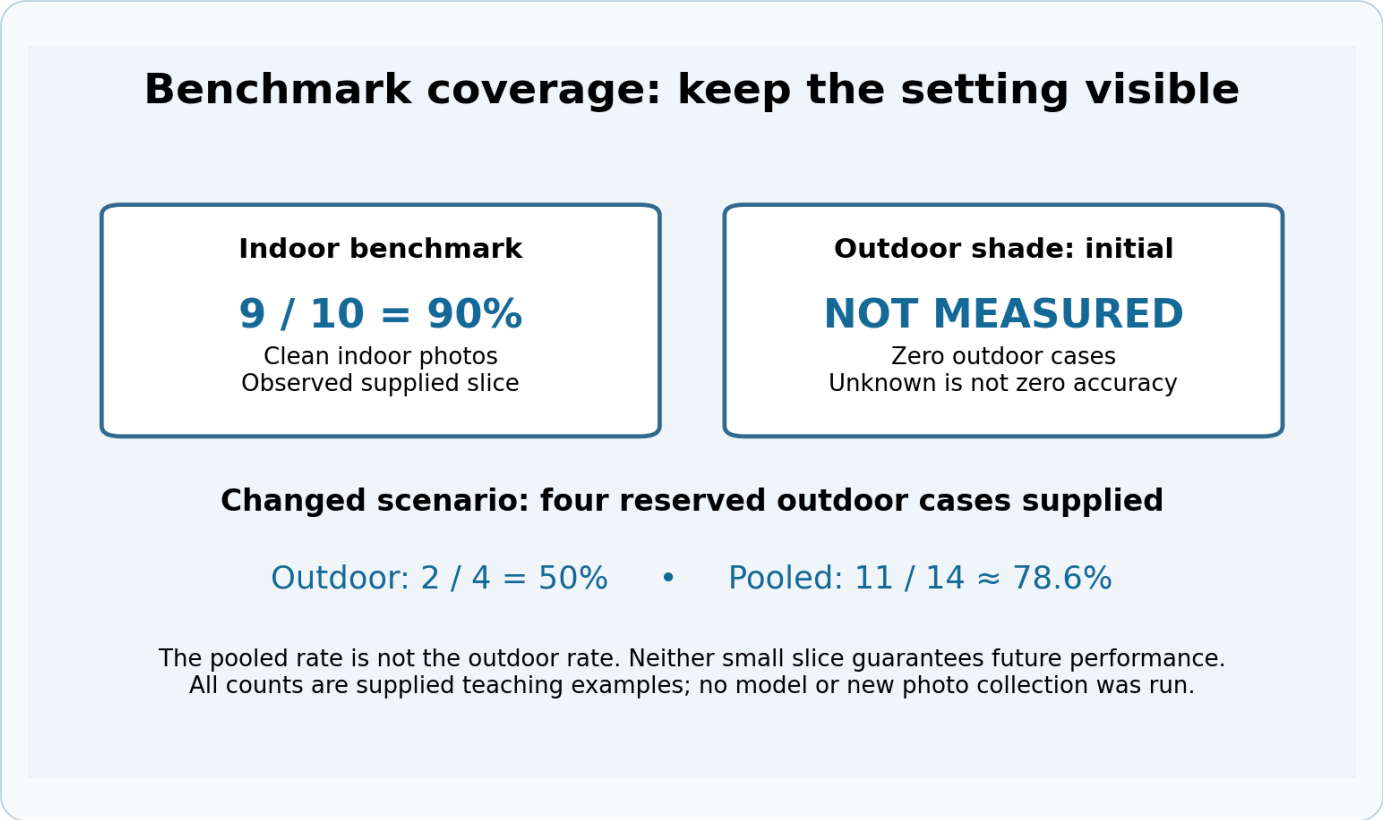
- A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

- Indoor **benchmark**: 9/10 correct. No outdoor cases are present. The same model is intended for both.

- Report the indoor **benchmark** with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/18

Benchmark conditions

	Correct	Total
Indoor benchmark	9	10
Added outdoor slice	2	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/18

Dataset and benchmark limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Name measured setting

All ten benchmark images are clean indoor photos.

2. Report the result

The observed indoor score is 90%.

3. Name missing evidence

Outdoor shade performance remains unmeasured.

CHANGE ONE THING

Add four reserved outdoor cases, of which two are correct.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.
- Which labelled step matches this detail? Outdoor shade performance remains unmeasured.
- 1. Report the result
- 2. Name measured setting
- 3. Name missing **evidence**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Name missing **evidence** Outdoor shade performance remains unmeasured. A **benchmark** represents particular tasks and collection conditions.

High performance on it cannot automatically establish fitness for a different deployment setting.

Changed-case question and optional hint

- Changed-case practice: Add four reserved outdoor cases, of which two are correct. Which conclusion is supported by the supplied facts?
- 1. Report the outdoor slice as 2/4=50%, separately from indoor results.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Report the outdoor slice as 2/4=50%, separately from indoor results. The new records finally measure the missing condition, with a small denominator.

Repair and retain the record

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

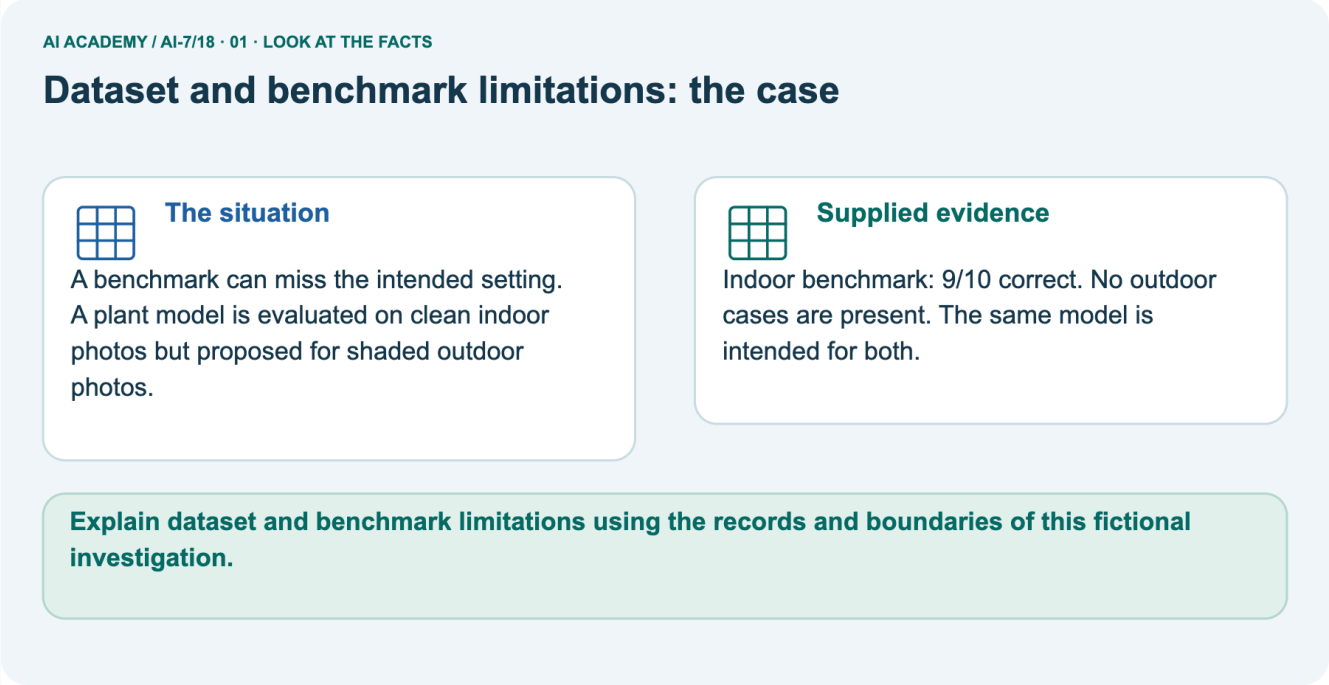
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Dataset and benchmark limitations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

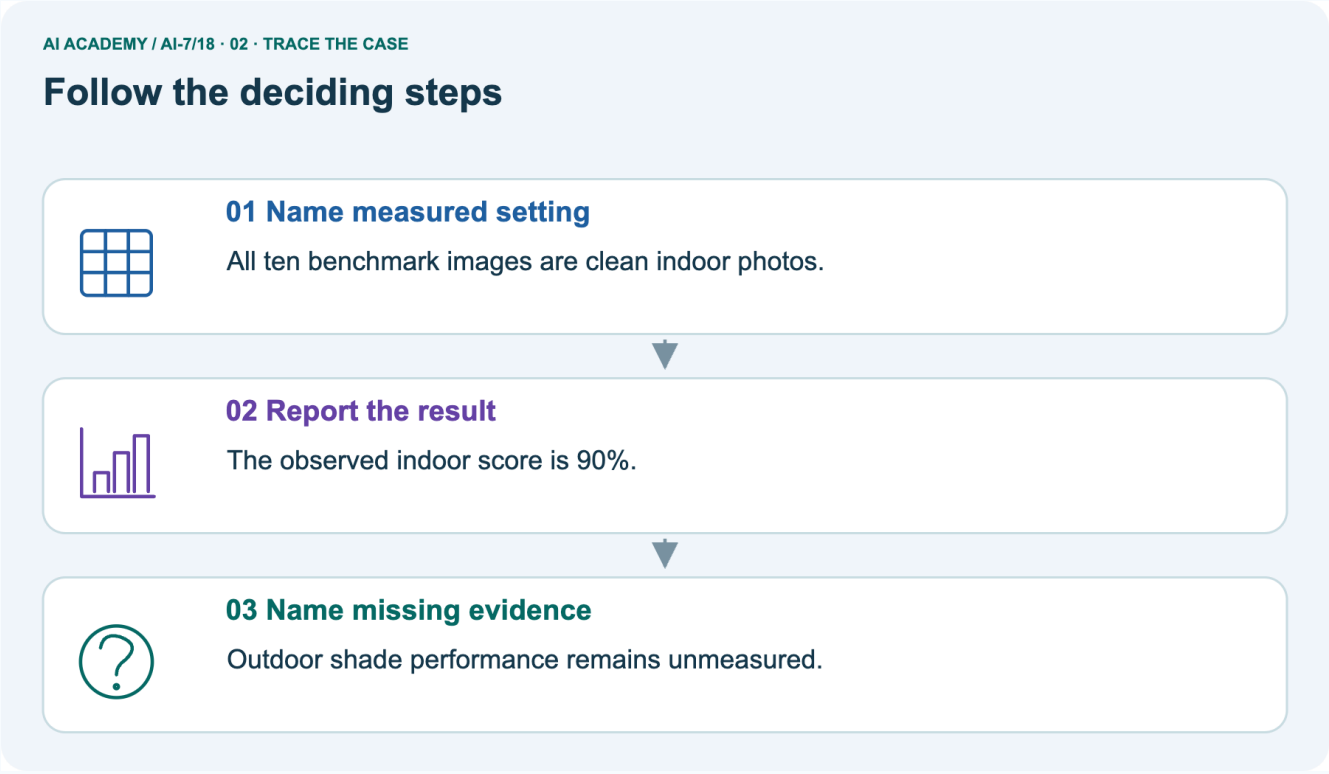
A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos. Supplied facts: Indoor **benchmark**: 9/10 correct. No outdoor cases are present.

The same model is intended for both.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



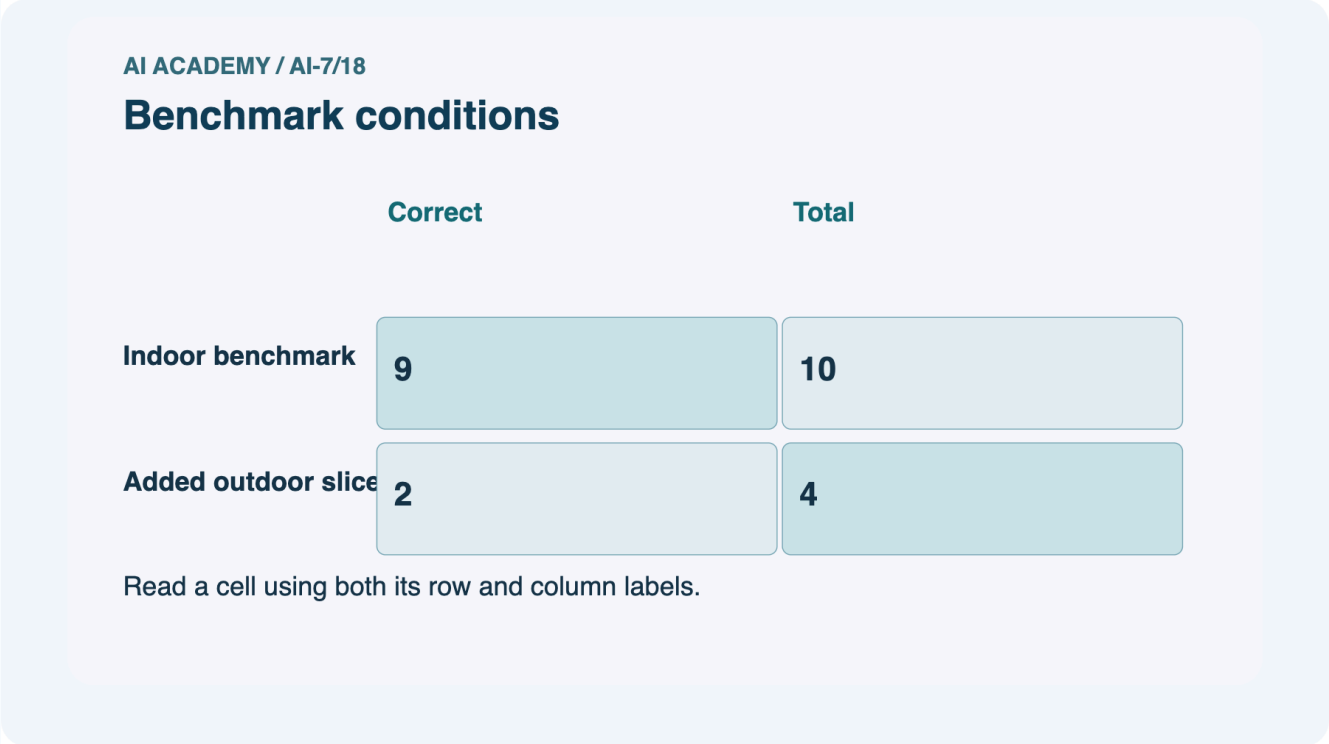
Follow the labelled picture with the matching explanation. Open the vector diagram

Name measured setting: All ten **benchmark** images are clean indoor photos. → Report the result: The observed indoor score is 90%. → Name missing **evidence**: Outdoor shade performance remains unmeasured.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Benchmark conditions



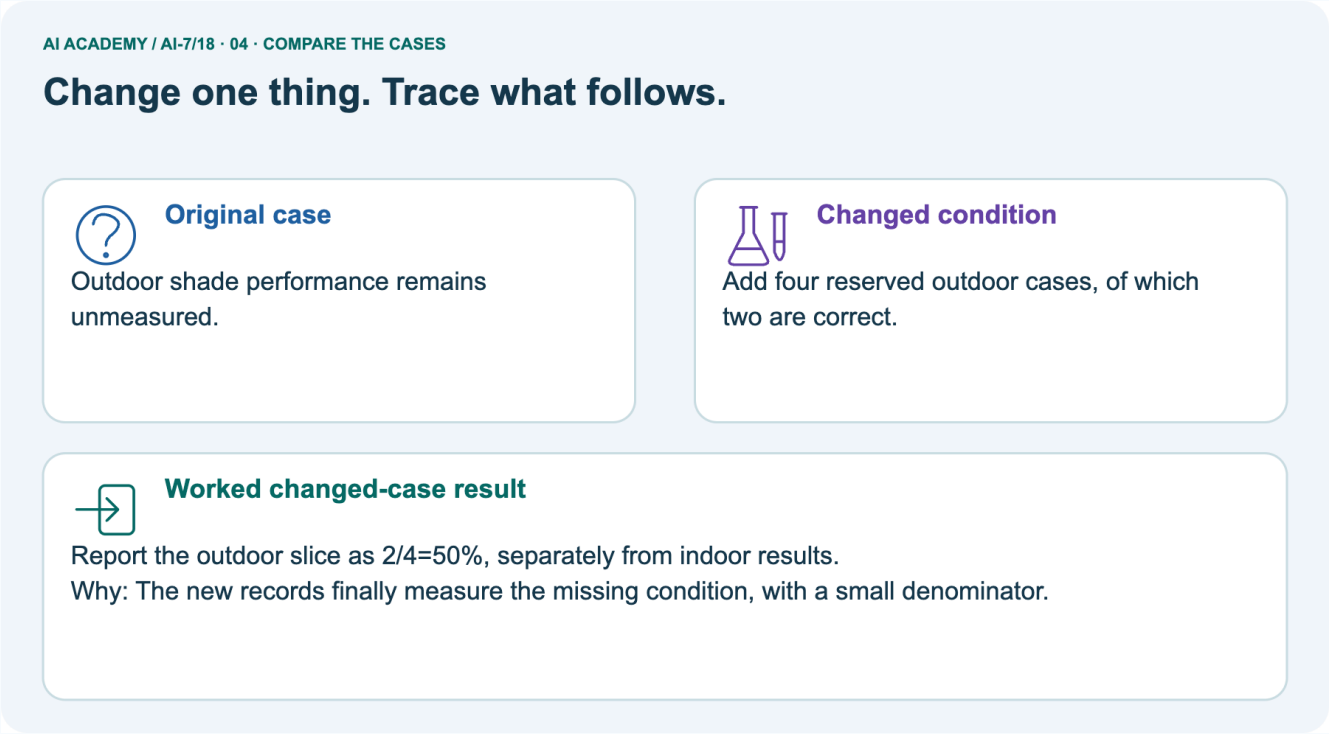
Follow the labelled picture with the matching explanation. Open the vector diagram

Benchmark conditions. Indoor **benchmark**: Correct = 9, Total = 10. Added outdoor slice: Correct = 2, Total = 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

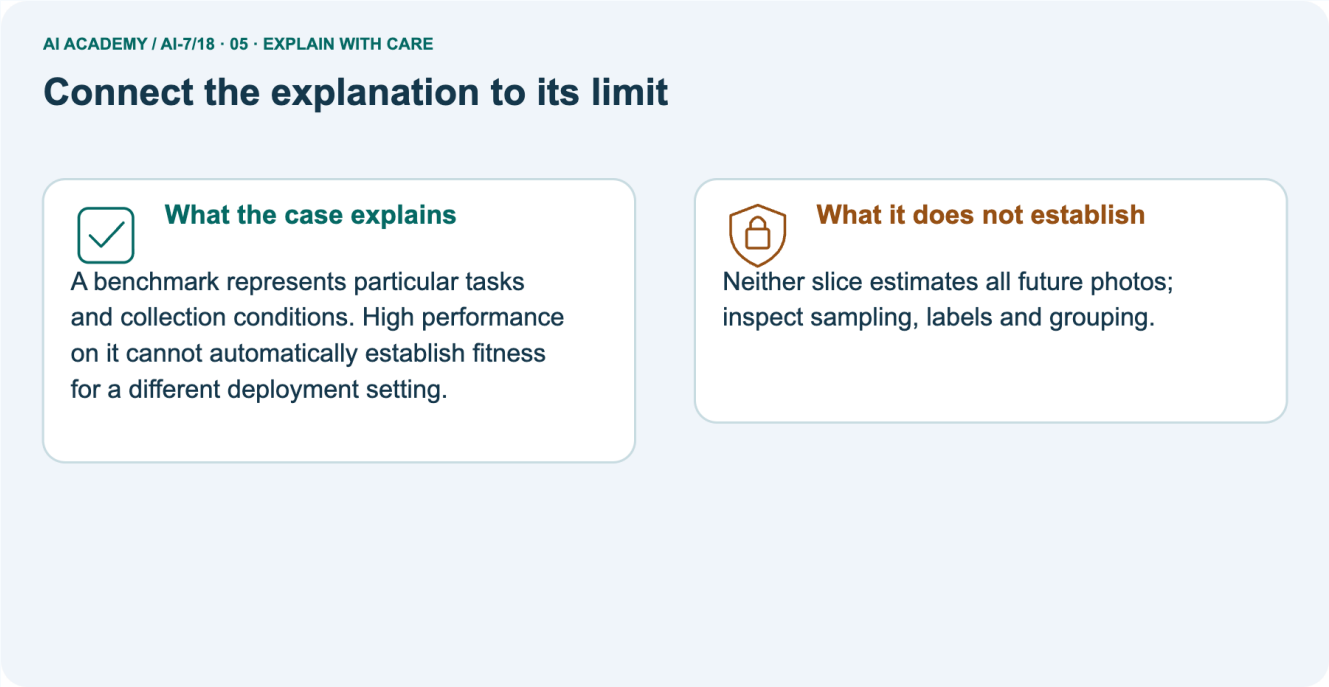
Original case: Outdoor shade performance remains unmeasured. Changed condition: Add four reserved outdoor cases, of which two are correct. Expected result: Report the outdoor slice as 2/4=50%, separately from indoor results.

Reason: The new records finally measure the missing condition, with a small denominator.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

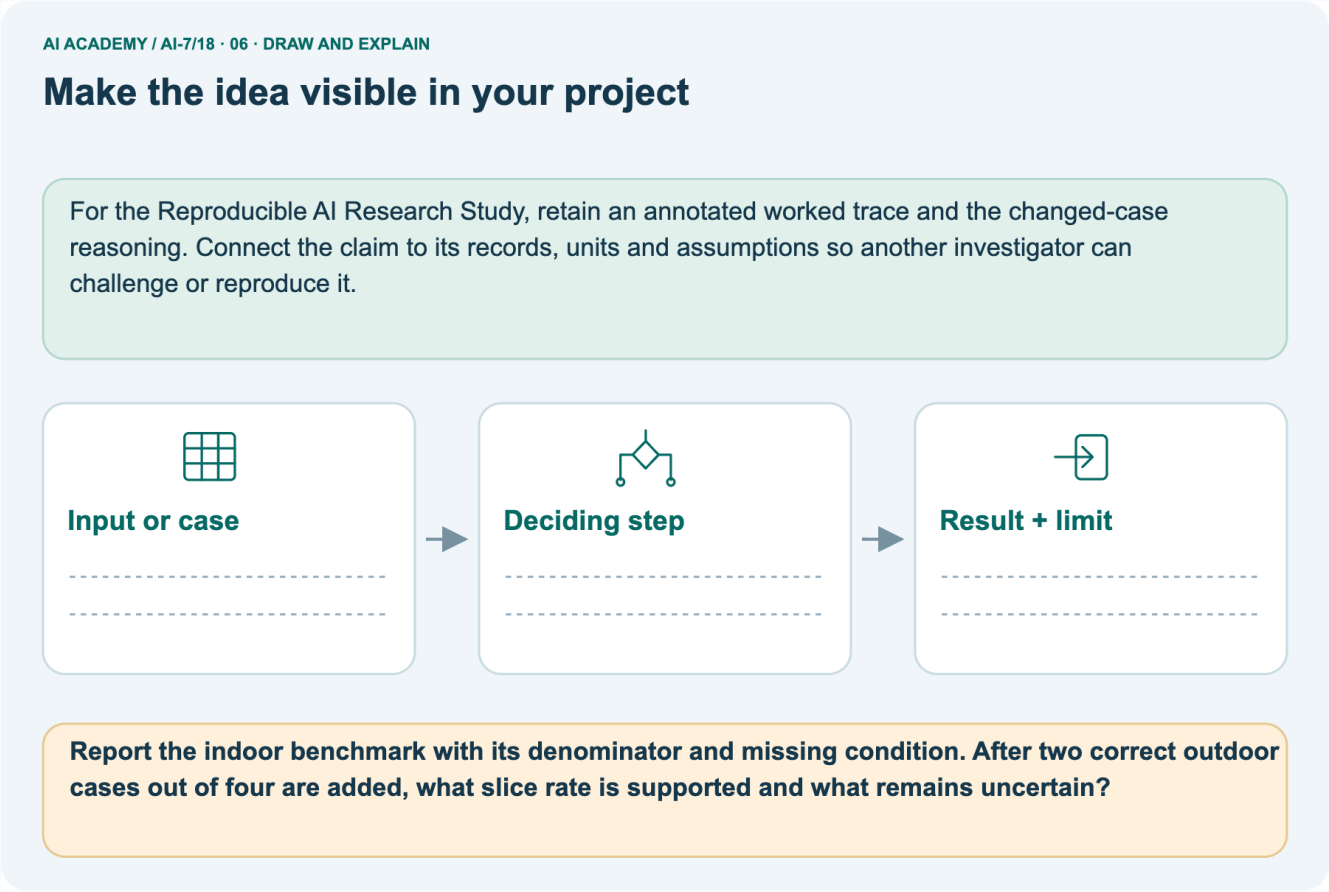
A **benchmark** represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

Limit: Neither slice estimates all future photos; inspect sampling, labels and grouping.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Report the indoor **benchmark** with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Earlier wording, alternatives and picture transcripts

01. Dataset and benchmark limitations: the case

02. Follow the deciding steps

03. Benchmark conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.